Eur J Appl Physiol (2004) 92: 352–359
DOI 10.1007/s00421-004-1110-9

ORIGINAL ARTICLE

**Travis W. Beck · Terry J. Housh · Glen O. Johnson
Joseph P. Weir · Joel T. Cramer · Jared W. Coburn
Moh H. Malek**

Mechanomyographic and electromyographic time and frequency domain responses during submaximal to maximal isokinetic muscle actions of the biceps brachii

Accepted: 16 March 2004 / Published online: 23 April 2004
© Springer-Verlag 2004

Abstract The purpose of this investigation was to determine the mechanomyographic (MMG) and electromyographic (EMG) amplitude and mean power frequency (MPF) versus torque relationships during isokinetic muscle actions of the biceps brachii. Twelve adults [mean (SD) age, 22.2 (2.7) years] performed submaximal to maximal isokinetic muscle actions of the dominant forearm flexors. Following determination of isokinetic peak torque (PT), the subjects randomly performed submaximal muscle actions in 20% increments from 20% to 80% PT. Polynomial regression analyses indicated linear increases in both MMG ($r^2 = 0.984$) and EMG ($r^2 = 0.988$) amplitude to 100% PT. There were no significant ($P > 0.05$) relationships, however, for MMG and EMG MPF versus isokinetic torque. The results demonstrated similar responses for MMG and EMG in both the time and frequency domains. These findings suggested that simultaneous examination of MMG and EMG amplitude and MPF may be useful for describing the unique motor control strategies that modulate dynamic torque production. Furthermore, the results indicated that dynamic muscle actions can be used when applying techniques that require a linear EMG amplitude versus torque relationship.

Keywords Isokinetic muscle actions · Mechanomyography · Motor control strategies

T. W. Beck (✉) · T. J. Housh · G. O. Johnson · J. W. Coburn
M. H. Malek
Department of Nutrition and Health Sciences,
Human Performance Laboratory,
Center for Youth Fitness and Sports Research,
University of Nebraska-Lincoln,
Lincoln, NE 68588-0229, USA
E-mail: tbeck@unlserve.unl.edu
Tel.: +1-402-4722690
Fax: +1-402-4720522

J. P. Weir
Applied Physiology Laboratory,
Program in Physical Therapy,
Des Moines University Osteopathic Medical Center,
Des Moines, IA 50312, USA

J. T. Cramer
Department of Kinesiology,
Exercise Science Research Laboratories,
The University of Texas at Arlington, Arlington,
TX 76019-0259, USA

Introduction

The mechanomyographic (MMG) signal records and quantifies the low-frequency lateral oscillations of active skeletal muscle fibers (Orizio 1993; Stokes 1993), and Gordon and Holbourn (1948) indicated that these oscillations reflect the "mechanical counterpart" of the motor unit electrical activity as measured by electromyography (EMG). Barry and Cole (1988) and Orizio (1993) have suggested that the lateral oscillations recorded as MMG are generated by (1) a gross lateral movement of the muscle at the initiation of a contraction that is generated by non-simultaneous activation of muscle fibers, (2) smaller subsequent lateral oscillations occurring at the resonant frequency of the muscle, and (3) dimensional changes of the active muscle fibers. The MMG signal, however, is influenced by many factors, including muscle temperature, stiffness, mass, intramuscular pressure, the viscosity of the intracellular and extracellular fluid media, and the firing rates of the active motor units (Marchetti et al. 1992; Orizio and Veicsteinas 1992; Orizio 1993; Stokes 1993; Orizio et al. 2003).

Simultaneous measurements of MMG and EMG have been used to examine the dissociation between the electrical and mechanical events (excitation-contraction coupling) that occurs with fatigue (Stokes and Dalton 1991) and to monitor factors related to electromechanical and phonomechanical delay (Petitjean et al. 1992). Recent investigations have also examined the MMG amplitude and frequency responses during maximal

concentric and eccentric isokinetic muscle actions (Cramer et al. 2002a, 2002b) as well as maximal and submaximal cycle ergometry (Stout et al. 1997; Housh et al. 2000). Clinically, MMG may be useful for examining neuromuscular disorders (Rhatigan et al. 1986), including cerebral palsy (Akataki et al. 1996), myotonic dystrophy (Orizio et al. 1997), cranio-mandibular disorders (L'Estrange et al. 1993), chronic and severe low back pain (Yoshitake et al. 2001), diaphragmatic fatigue (Petitjean and Bellemare 1994), skeletal muscle atrophy (Marchetti et al. 1974), and as a control mechanism for externally powered prosthesis (Barry et al. 1986).

The MMG signal summates the activity from individual motor units, and it has been suggested that the time and frequency domains of the MMG signal may provide information regarding the motor control strategies (motor unit recruitment and firing rate) of various muscles during both isometric and dynamic muscle actions (Orizio et al. 1989, 1990, 2003; Dalton and Stokes 1991; Orizio 1993; Akataki et al. 2003). Specifically, it has been shown that MMG amplitude is related to motor unit recruitment, while the frequency domain may provide information regarding motor unit firing rate (Orizio 1993; Orizio et al. 2003). Thus, simultaneous examination of the time and frequency domains of the MMG signal may be useful for describing the differences that exist in the motor control strategies that modulate force production during isometric versus dynamic muscle actions (Kossev and Christova 1998; Linnamo et al. 2003). No studies, however, have examined the MMG and EMG responses during submaximal to maximal, isokinetic muscle actions. Therefore, the purpose of the present study was to examine the MMG and EMG amplitude and mean power frequency (MPF) versus torque relationships during submaximal to maximal isokinetic muscle actions of the biceps brachii.

Methods

Subjects

Twelve adults [6 males and 6 females, mean (SD) age, 22.2 (2.7) years] volunteered to participate in this investigation. The study was approved by the University Institutional Review Board for Human Subjects, and all subjects completed a health history questionnaire and signed a written informed consent document prior to testing. The experiments performed in this investigation comply with the current laws of the United States of America.

Orientation session

The orientation session familiarized the subject with the isokinetic testing procedures. The concentric isokinetic muscle actions of the dominant forearm flexors (based on throwing preference) were performed on a calibrated Cybex II dynamometer. Following assessment of isokinetic peak torque (PT) at 30° s^{-1}, the subjects practiced five submaximal isokinetic muscle actions. Verbal feedback regarding torque production was provided after each isokinetic muscle action.

Isokinetic testing

After a minimum rest period of 24 h following the orientation session, each subject performed the isokinetic testing. For all muscle actions, the subjects were positioned and stabilized in accordance with the Cybex II instruction manual (Cybex 1980). The subjects were tested in a supine position and used a neutral handgrip. Prior to the isokinetic testing session, the subjects performed a warm-up of five isokinetic muscle actions at 30° s^{-1}. They were instructed to provide an effort corresponding to approximately 50% of their maximum for each muscle action. Following the warm-up and a rest period of 2 min, the subjects performed two maximal concentric isokinetic muscle actions at 30° s^{-1} with the highest torque output selected as the PT value. After PT was determined, the subjects performed a series of randomly ordered submaximal muscle actions, in 20% increments from 20% to 80% PT. Trials were repeated if the actual submaximal torque was not within 5% of the calculated value. A 2-min rest was allowed between each muscle action. Following the submaximal isokinetic muscle actions, two additional maximal efforts were performed to determine if the testing had affected PT.

Mechanomyography

The MMG signal was detected by a piezoelectric crystal contact sensor (Hewlett-Packard, 21050A, bandwidth 0.02–2,000 Hz; Andover, Mass., USA) placed over the belly of the biceps brachii muscle (Orizio et al. 1989, 1990) (Fig. 1). A stabilizing ring, double-sided foam tape, and microporous tape were used to ensure consistent contact pressure of the MMG sensor (Bolton et al. 1989).

Electromyography

Bipolar surface (7.62 cm center to center) electrode (Quinton Quick prep silver/silver chloride) arrangements were placed on the dominant arm over the mid-portion of the biceps brachii muscle with the reference electrode placed over the anterior distal end of the forearm between the styloid processes of the radius and ulna (Fig. 1). The interelectrode distance was selected to accommodate placement of the MMG sensor between the active EMG electrodes (Ebersole et al. 2002). Interelectrode impedance was kept below

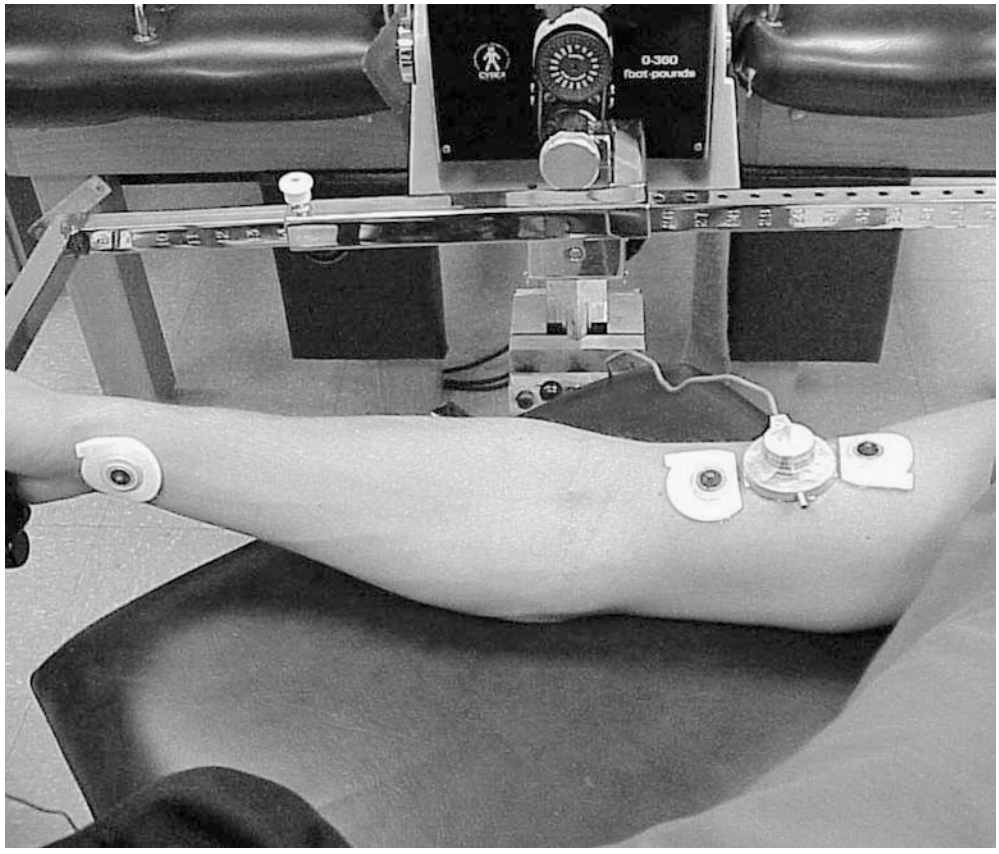

Fig. 1 Example of the mechanomyography sensor and electromyography electrode placements

2,000 Ω by careful skin abrasion. The EMG signal was amplified (gain: ×1,000) using a differential amplifier (EMG 100, bandwidth 1.0–5,000 Hz; Biopac Systems, Santa Barbara, Calif., USA).

Signal processing

The raw MMG and EMG signals were digitized at 1,000 Hz and stored in a personal computer (Macintosh 7100/80 AV Power PC, Apple Computer, Cupertino, Calif., USA) for subsequent analysis. All signal processing was performed using custom programs written with LabVIEW programming software (version 6.1; National Instruments, Austin, Tex., USA). The MMG and EMG signals were bandpass filtered (fourth-order Butterworth) at 5–100 Hz and 10–500 Hz, respectively, and the amplitude (root-mean-square, rms) and mean power frequency (MPF) values for the isokinetic muscle actions were calculated for a time period that corresponded to a 50° range of motion from approximately 110° to 160° of forearm flexion. Thus, at 30° s^{-1}, the amplitude and MPF for 1.67 s were calculated (Fig. 2). This portion of the range of motion was selected to avoid the acceleration and deceleration phases of movement which are typical of isokinetic dynamometers (Brown et al. 1995). For the MPF analyses, each data segment was processed with a Hamming window and discrete Fourier transformations (DFT). The MPF was selected to represent the power spectrum on the basis of the recommendations of Diemont et al. (1988) and was calculated as described by Kwatny et al. (1970).

Statistical analysis

Torque, MMG amplitude, MMG MPF, EMG amplitude, and EMG MPF values were determined for all muscle actions of each test. The average MMG amplitude, MMG MPF, EMG amplitude and EMG MPF in relation to %PT were examined using polynomial regression models (linear, quadratic, cubic; SPSS software program, Chicago, Ill., USA). Using X = %PT, Y = MMG amplitude, MMG MPF, EMG amplitude or EMG MPF, and a_0, a_1, a_2, and a_3 = statistically determined regression coefficients, these models are:

$$Y = a_0 + a_1X \text{ (linear model)}$$

$$Y = a_0 + a_1X + a_2X^2 \text{ (quadratic model)}$$

$$Y = a_0 + a_1X + a_2X^2 + a_3X^3 \text{ (cubic model)}$$

The statistical significance ($P \leq 0.05$) for the increment in the proportion of the variance that would be accounted for by a higher-degree polynomial was determined using the following F-test (Pedhazur 1997):

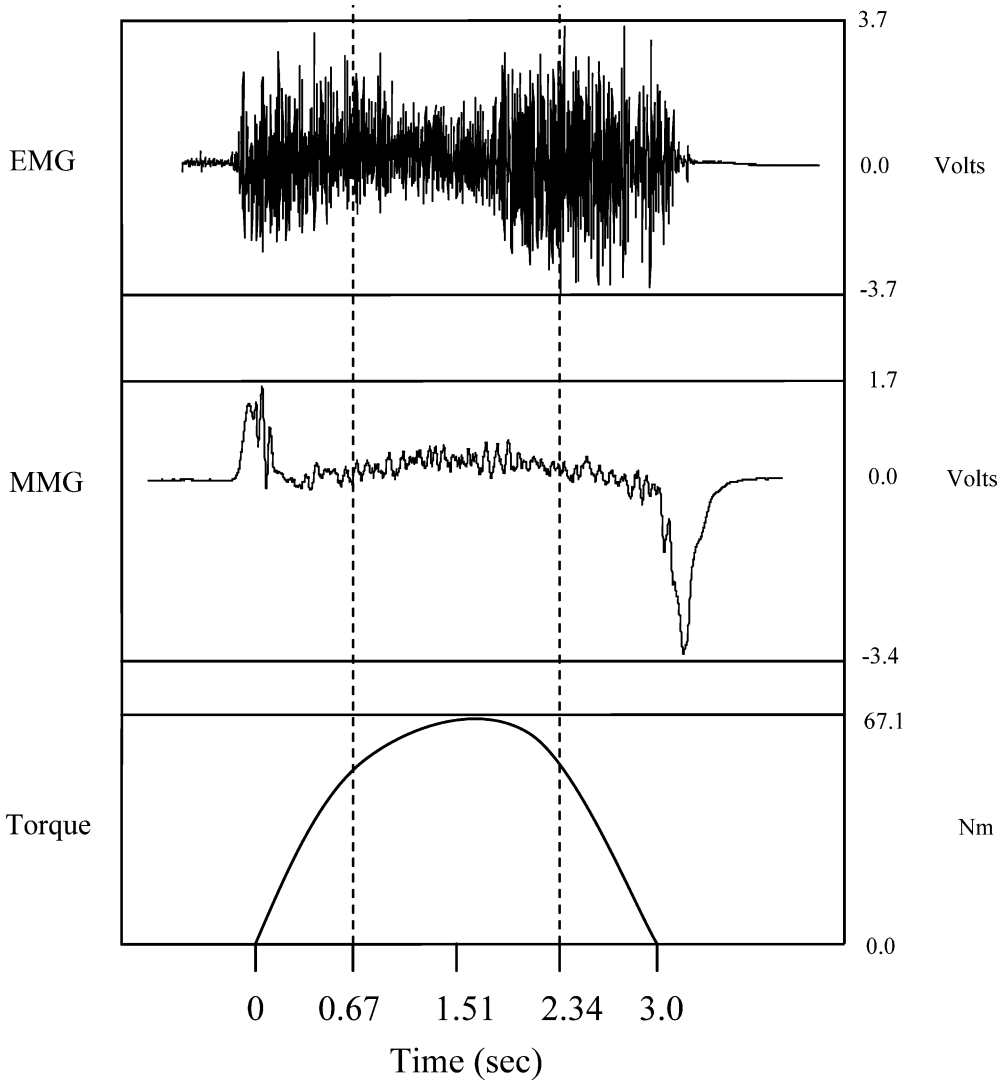

Fig. 2 Example of the raw mechanomyographic (*MMG*)and electromyographic (*EMG*)signals from the biceps brachii and the isokinetic torque production curve. The portions of the MMG, EMG, and torque signals between the *dashed vertical lines* were selected for analysis

$$F = \frac{\left(R_2^2 - R_1^2\right)/(K_2 - K_1)}{\left(1 - R_2^2\right)/(n - K_2 - 1)}$$

where n is the number of data points, K_2 is the number of predictors from the larger R^2 and K_1 is the number of predictors from the smaller R^2. Paired t-tests were used to determine if there were differences between the means for PT values measured prior to and following the submaximal isokinetic muscle actions. An alpha of 0.05 was considered statistically significant for all comparisons.

Results

Torque

The mean (SEM) isokinetic PT was 44.2 (6.5) Nm. Following the submaximal muscle actions, the mean (SEM) isokinetic PT was 43.6 (6.7) Nm. There was no significant ($P > 0.05$) difference between the PT values measured prior to and following the submaximal isokinetic muscle actions.

Mechanomyography

As shown in Fig. 3, the MMG amplitude (mV rms) versus %PT relationship was best fit with a linear model ($r^2 = 0.984$). There was no significant ($P > 0.05$) relationship (Fig. 4) between MMG MPF (Hz) and %PT.

Electromyography

As shown in Fig. 5, the EMG amplitude (μV rms) versus %PT relationship was best fit with a linear model ($r^2 = 0.988$). There was no significant ($P > 0.05$) relationship (Fig. 6) between EMG MPF (Hz) and %PT.

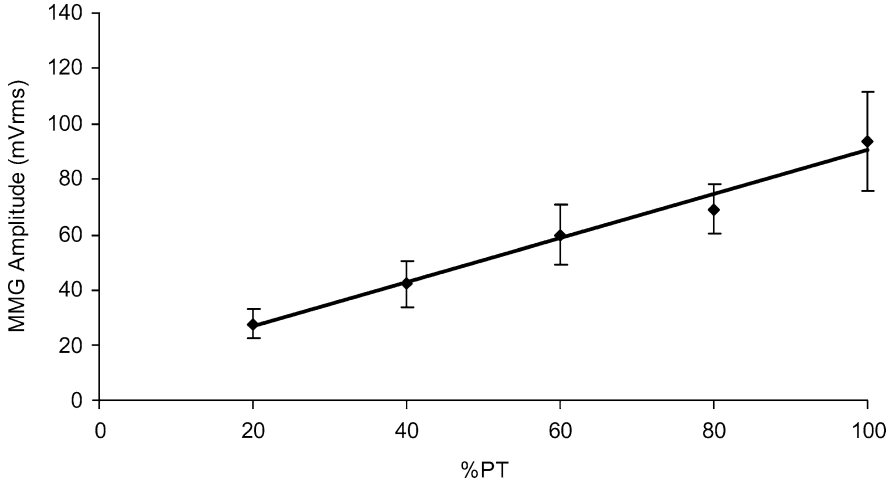

Fig. 3 MMG amplitude (mV rms) as a function of the percentage peak torque (%PT) during isokinetic muscle actions of the biceps brachii. Polynomial regression analyses indicated that the relationship was best fit with a linear ($r^2 = 0.984$) model. Values are mean (SEM)

Discussion

Mechanomyography

The amplitude of the MMG signal is determined by the number of active motor units and their firing frequencies (Orizio 1993; Orizio et al. 2003). In addition, it has been suggested that the frequency content of the MMG signal may qualitatively reflect the global firing rate of the unfused activated motor units (Orizio 1993; Bichler 2000; Akataki et al. 2001, 2003; Bichler and Celichowski 2001; Orizio et al. 2003). Thus, simultaneous examination of the time and frequency domains of the MMG signal may provide information regarding the unique motor control strategies (recruitment and firing rate) that modulate force production in various muscles (Orizio et al. 1989, 1990; Zhang et al. 1992; Kossev and Christova 1998; Shinohara et al. 1998; Madeleine et al. 2001; Linnamo et al. 2003; Akataki et al. 2003).

Previous studies (Dalton and Stokes 1991; Petitjean et al. 1992) have demonstrated linear MMG amplitude versus torque relationships during submaximal concentric and eccentric muscle actions of the biceps brachii. The present study differed from the experiments of Dalton and Stokes (1991) and Petitjean et al. (1992) in that it included maximal as well as submaximal isokinetic muscle actions at a single velocity (30° s^{-1}) over a range of 20–100% PT. Dalton and Stokes (1991) hypothesized that the linear relationship between MMG amplitude and concentric torque was due to an increase in the number of active motor units as torque increased. Furthermore, Orizio et al. (1989, 1993, 2003) have reported that MMG amplitude increased to the end of motor unit recruitment during voluntary, incremental step and ramp isometric muscle actions of the biceps brachii as well as electrically stimulated ramp contractions of the cat gastrocnemius. Therefore, the linear ($r^2 = 0.984$) MMG amplitude versus %PT relationship found in the present study suggested that motor unit

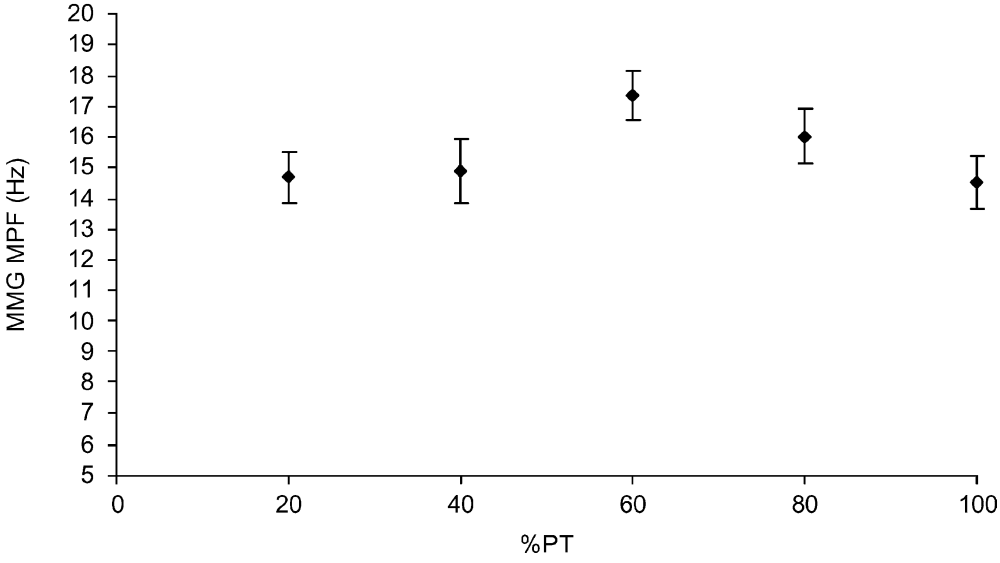

Fig. 4 MMG mean power frequency (*MPF*, Hz) as a function of %PT during isokinetic muscle actions of the biceps brachii. Polynomial regression analyses indicated that there was no significant ($P > 0.05$) relationship between the variables. Values are mean (SEM)

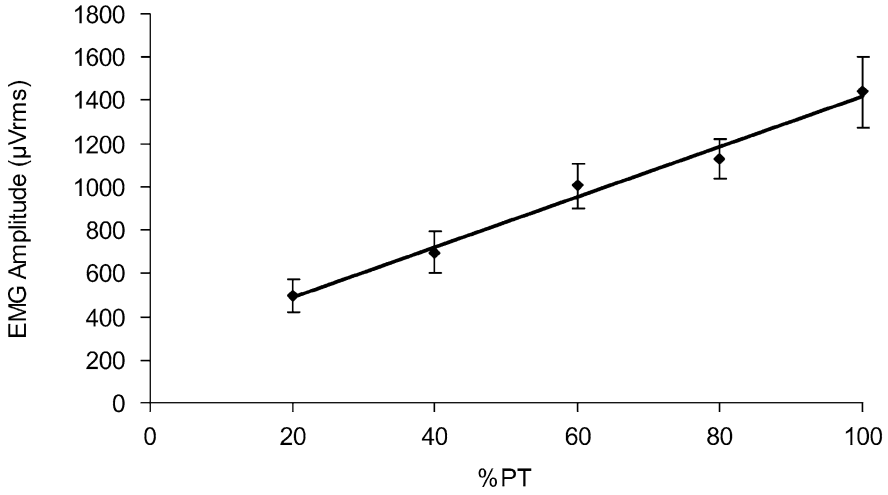

Fig. 5 EMG amplitude (μV rms) as a function of %PT during isokinetic muscle actions of the biceps brachii. Polynomial regression analyses indicated that the relationship was best fit with a linear ($r^2 = 0.988$) model. Values are mean (SEM)

recruitment continued throughout the entire torque range from 20% to 100% PT.

The lack of change in MMG MPF suggested that the global motor unit firing rate in the biceps brachii remained stable across the isokinetic torque levels. This was consistent with the findings of Kossev and Christova (1998) who stated "...the gradation of muscle force during concentric movements could be assigned more to recruitment than to the MUs (motor units) rate coding" (p 253). In addition, Linnamo et al. (2003) found that during concentric muscle actions, the mean spike frequency detected with intramuscular electrodes increased from 20% to 40% PT, and then remained stable from 40% to 80% PT (Linnamo et al. 2003). The current findings as well as those of Kossev and Christova (1998) and Linnamo et al. (2003) are in contrast to the typical motor control strategy for isometric muscle actions, in which torque production is modulated by concurrent increases in motor unit recruitment and firing rates to about 50–90% of the maximum voluntary contraction (MVC), and then further increased to MVC by rate coding alone (Kukulka and Clamann 1981; Enoka and Fuglevand 2001). Therefore, the results for MMG amplitude and MPF in the present study, in conjunction with the results of previous studies (Kossev and Christova 1998; Linnamo et al. 2003) suggested that recruitment, with little change in the global motor unit firing rate, may be the primary motor control strategy to increase concentric isokinetic torque in the biceps brachii.

Electromyography

The amplitude of the EMG signal reflects muscle activation and is determined by the number of recruited motor units and their firing rates (Basmajian and De Luca 1985). Furthermore, it has been suggested that the EMG power spectrum reflects muscle fiber action potential conduction velocity, which, in turn, may provide information about fiber-type recruitment patterns (Westbury and Shaughnessy 1987). Specifically, the EMG center frequency (mean or median) tends to be lower for low threshold, slow-twitch motor units, than high threshold, fast-twitch motor units.

In the present study, there was a highly linear relationship ($r^2 = 0.988$) between EMG amplitude and concentric isokinetic torque. Previous studies (Komi 1973;

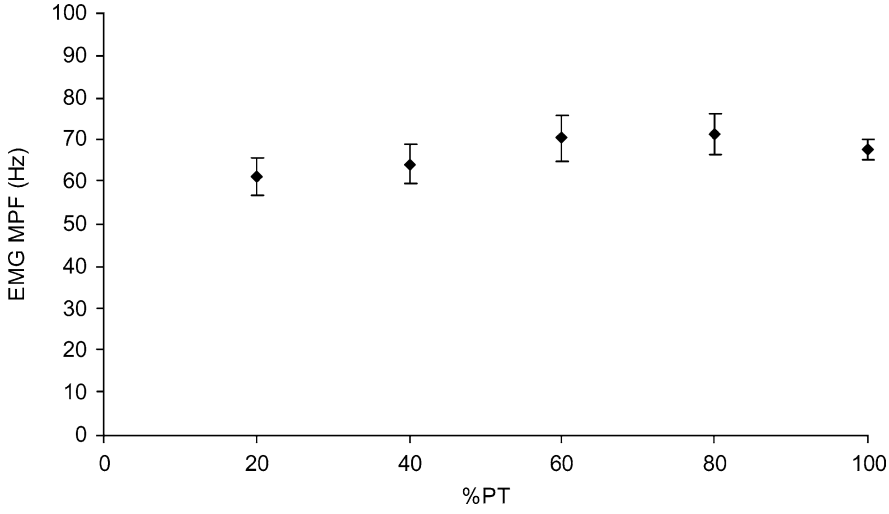

Fig. 6 EMG MPF (Hz) as a function of %PT during isokinetic muscle actions of the biceps brachii. Polynomial regression analyses indicated that there was no significant ($P > 0.05$) relationship between the variables. Values are mean (SEM)

Dalton and Stokes 1991; Petitjean et al. 1992) have also reported linear relationships between EMG amplitude and torque during dynamic muscle actions of the biceps brachii. For example, Komi (1973) reported a linear relationship during submaximal to maximal concentric isokinetic muscle actions of the biceps brachii performed at high and low velocities. The linear EMG amplitude versus torque relationship for dynamic muscle actions differs from the curvilinear relationship often found for isometric muscle actions (Lawrence and DeLuca 1983; Woods and Bigland-Ritchie 1983). Woods and Bigland-Ritchie (1983) suggested that a curvilinear relationship during isometric muscle actions may reflect further increases in the firing rates of slow-twitch motor units that are already responding at tetanic frequencies and producing their maximum force. Theoretically, increases in the discharge rates of these motor units would cause an increase in EMG amplitude without a corresponding increase in torque production, resulting in a curvilinear EMG amplitude versus isometric torque relationship (Woods and Bigland-Ritchie 1983). Therefore, the linear increase in EMG amplitude in the present study is consistent with the results for MMG amplitude and MPF and suggested that motor unit recruitment, rather than increases in firing rate may be the primary motor control strategy for increasing concentric isokinetic torque in the biceps brachii.

The present findings also suggest that techniques that require a linear EMG amplitude versus torque relationship such as the "efficiency of electrical activity" (EEA) procedure of DeVries (1968) can be applied to dynamic muscle actions. Specifically, DeVries (1968) proposed that a decrease in the linear slope coefficient for the EMG amplitude versus isometric torque relationship indicated a reduction in the electrical activity that was required to produce a given level of torque and an improvement in muscle function. The current findings suggest that bipolar EMG methodology and dynamic muscle actions can be used when applying the EEA technique to examine issues such as the time course of neural versus hypertrophic contributions to training-induced strength gains and the mechanisms underlying the cross-training effect.

Previous studies (Moritani and Muro 1987; Solomonow et al. 1990) have suggested that increases in EMG center frequency (mean or median) with increases in isometric torque production may reflect recruitment of larger motor units, which have higher action potential conduction velocities than smaller motor units. Recently, however, Masuda et al. (2001) provided evidence that the relationship between EMG center frequency and action potential conduction velocity does not hold for dynamic muscle actions by showing no change in EMG median frequency from the vastus lateralis muscle, but increases in action potential conduction velocity with increases in torque during isokinetic leg extensions. In the present study, EMG MPF remained stable from 20% to 100% PT. These findings supported the hypothesis of Masuda et al. (2001) and suggested that EMG center frequency may not reflect action potential conduction velocity during dynamic muscle actions.

In summary, there were linear increases for MMG and EMG amplitude with isokinetic torque, but no change for MMG or EMG MPF. The similar results for MMG and EMG in the time and frequency domains suggested that for the biceps brachii, dynamic torque production may be modulated more by recruitment than by increases in the global motor unit firing rate. These findings were in contrast to the typical motor control strategy for isometric muscle actions, but were similar to the results from previous studies of dynamic muscle actions (Kossev and Christova 1998; Linnamo et al. 2003). Therefore, simultaneous analyses of MMG and EMG amplitude and MPF may provide information regarding the unique motor control strategies used to modulate torque production during dynamic muscle actions.

References

Akataki K, Mita K, Itoh K, Suzuki N, Watakabe M (1996) Acoustic and electrical activities during voluntary isometric contraction of biceps brachii muscle in patients with spastic cerebral palsy. Muscle Nerve 19:1252–1257

Akataki K, Mita K, Watakabe M, Itoh K (2001) Mechanomyogram and force relationship during voluntary isometric ramp contractions of the biceps brachii muscle. Eur J Appl Physiol 84:19–25

Akataki K, Mita K, Watakabe M, Itoh K (2003) Mechanomyographic responses during voluntary ramp contractions of the human first dorsal interosseous muscle. Eur J Appl Physiol 89:520–525

Barry DT, Cole NM (1988) Fluid mechanics of muscle vibrations. Biophys J 53:899–905

Barry DT, Leonard JA, Gitter AJ, Ball RD (1986) Acoustic myography as a control signal for externally powered prosthesis. Arch Phys Med Rehabil 67:267–269

Basmajian JV, De Luca CJ (1985) Muscles alive: their functions revealed by electromyography, 5th edn. Williams and Wilkins, Baltimore, MD

Bichler E (2000) Mechanomyograms recorded during evoked contractions of single motor units in the rat medial gastrocnemius muscle. Eur J Appl Physiol 83:310–319

Bichler E, Celichowski J (2001) Mechanomyographic signals generated during unfused tetani of single motor units in the rat medial gastrocnemius muscle. Eur J Appl Physiol 85:513–520

Bolton CF, Parkes A, Thompson TR, Clark MR, Sterne CJ (1989) Recording sound from human skeletal muscle: technical and physiological aspects. Muscle Nerve 12:126–134

Brown LE, Whitehurst M, Gilbert R, Buchhalter DN (1995) The effect of velocity and gender on load range during knee extension and flexion exercise on an isokinetic device. J Orthop Sports Phys Ther 21:107–112

Cramer JT, Housh TJ, Weir JP, Johnson GO, Berning JM, Perry SR, Bull AJ (2002a) Mechanomyographic and electromyographic amplitude and frequency responses from the superficial quadriceps femoris muscles during maximal, eccentric isokinetic muscle actions. Electromyogr Clin Neurophysiol 42:337–346

Cramer JT, Housh TJ, Weir JP, Johnson GO, Ebersole KT, Perry SR, Bull AJ (2002b) Power output, mechanomyographic, and electromyographic responses to maximal, concentric, isokinetic muscle actions in men and women. J Strength Cond Res 16:399–408

Cybex (1980) Isolated joint testing and exercise—a handbook for using CYBEX II and the U.B.X.T. (owner's manual). Cybex, Bay Shore, NY, pp 49–51

Dalton PA, Stokes MJ (1991) Acoustic myography reflects force changes during dynamic concentric and eccentric contractions of the human biceps brachii muscle. Eur J Appl Physiol 63:412–416

DeVries HA (1968) "Efficiency of electrical activity" as a physiological measure of the functional state of muscle tissue. Am J Phys Med 47:10–22

Diemont B, Figini MM, Orizio C, Perini R, Veicsteinas A (1988) Spectral analysis of muscular sound at low and high contraction level. Int J Biomed Comput 23:161–175

Ebersole KT, Housh TJ, Johnson GO, Perry SR, Bull AJ, Cramer JT (2002) Mechanomyographic and electromyographic responses to unilateral isometric training. J Strength Cond Res 16:192–201

Enoka RM, Fuglevand AJ (2001) Motor unit physiology: some unresolved issues. Muscle Nerve 24:4–17

Gordon G, Holbourn AHS (1948) The sounds from single motor units in a contracting muscle. J Physiol (Lond) 107:456–464

Housh TJ, Perry SR, Bull AJ, Johnson GO, Ebersole KT, Housh DJ, DeVries HA (2000) Mechanomyographic and electromyographic responses during submaximal cycle ergometry. Eur J Appl Physiol 83:381–387

Komi PV (1973) Relationship between muscle tension, EMG and velocity of contraction under concentric and eccentric work. In: Desmedt J (eds) New developments in electromyography and clinical neurophysiology, vol 1. Karger, Basel, pp 596–606

Kossev A, Christova P (1998) Discharge pattern of human motor units during dynamic concentric and eccentric contractions. Electroencephalogr Clin Neurophysiol 109:245–255

Kukulka CG, Clamann HP (1981) Comparison of the recruitment and discharge properties of motor units in human brachial biceps and adductor pollicis during isometric contractions. Brain Res 219:45–55

Kwatny E, Thomas DH, Kwatny HG (1970) An application of signal processing techniques to the study of myoelectric signals. IEEE Trans Biomed Eng 17:303–312

Lawrence JH, De Luca CJ (1983) Myoelectric signal versus force relationship in different human muscles. J Appl Physiol 54:1653–1659

L'Estrange PR, Rowell J, Stokes MJ (1993) Acoustic myography in the assessment of human masseter muscle. J Oral Rehabil 20:353–362

Linnamo V, Moritani T, Nicol C, Komi PV (2003) Motor unit activation patterns during isometric, concentric and eccentric actions at different force levels. J Electromyogr Kinesiol 13:93–101

Madeleine P, Bajaj P, Søgaard K, Arendt-Nielsen L (2001) Mechanomyography and electromyography force relationships during concentric, isometric and eccentric contractions. J Electromyogr Kinesiol 11:113–121

Marchetti M, Salleo A, Figura F, Del Gaudio V (1974) Electromyographic and phonomyographic patterns in muscle atrophy in man. In: Nelson RC, Morehouse CA (eds) Biomechanics IV. University Park Press, Baltimore, MD, pp 388–393

Marchetti M, Felici F, Bernardi M, Minasi P, Di Filippo L (1992) Can evoked phonomyography be used to recognize fast and slow twitch muscle in man? Int J Sports Med 13:65–68

Masuda T, Kizuka T, Yong Zhe J, Yamada H, Saitou K, Sadoyama T, Okada M (2001) Influence of contraction force and speed on muscle fiber conduction velocity during dynamic voluntary exercise. J Electromyogr Kinesiol 11:85–94

Moritani T, Muro M (1987) Motor unit activity and surface electromyogram power spectrum during increasing force of contraction. Eur J Appl Physiol 56:260–265

Orizio C (1993) Muscle sound: bases for the introduction of a mechanomyographic signal in muscle studies. Crit Rev Biomed Eng 21:201–243

Orizio C, Veicsteinas A (1992) Soundmyogram analysis during sustained maximal voluntary contraction in sprinters and long distance runners. Int J Sports Med 13:594–599

Orizio C, Perini R, Veicsteinas A (1989) Muscular sound and force relationship during isometric contraction in man. Eur J Appl Physiol 58:528–533

Orizio C, Perini R, Diemont B, Figini MM, Veicsteinas A (1990) Spectral analysis of muscular sound during isometric contraction of biceps brachii. J Appl Physiol 68:508–512

Orizio C, Solomonow M, Baratta R, Veicsteinas A (1993) Influence of motor unit recruitment and firing rate on the sound myogram and EMG characteristics in cat gastrocnemius. J Electromyogr Kinesiol 2:232–241

Orizio C, Esposito F, Sansone V, Parrinello G, Meola G, Veicsteinas A (1997) Muscle surface mechanical and electrical activities in myotonic dystrophy. Electromyogr Clin Neurophysiol 37:231–239

Orizio C, Gobbo M, Diemont B, Esposito F, Veicsteinas A (2003) The surface mechanomyogram as a tool to describe the influence of fatigue on biceps brachii motor unit activation strategy. Historical basis and novel evidence. Eur J Appl Physiol 90:326–336

Pedhazur EJ (1997) Multiple regression in behavioral research: explanation and prediction, 3rd edn. Harcourt Brace, Fort Worth, TX, pp 520–521

Petitjean M, Bellemare F (1994) Phonomyogram of the diaphragm during unilateral and bilateral phrenic nerve stimulation and changes with fatigue. Muscle Nerve 17:1201–1209

Petitjean M, Maton B, Cnockaert J-C (1992) Evaluation of human dynamic contraction by phonomyography. J Appl Physiol 73:2567–2573

Rhatigan BA, Mylrea KC, Lonsdale E, Stern LZ (1986) Investigation of sounds produced by healthy and diseased human muscular contraction. IEEE Trans Biomed Eng 33:967–971

Shinohara M, Kouzaki M, Yoshihisa T, Fukunaga T (1998) Mechanomyogram from the different heads of the quadriceps muscle during incremental knee extension. Eur J Appl Physiol 78:289–295

Solomonow M, Baten C, Smit J, Baratta R, Hermens H, D'Ambrosia R, Shoji H (1990) Electromyogram power spectra frequencies associated with motor unit recruitment strategies. J Appl Physiol 68:1177–1185

Stokes MJ (1993) Acoustic myography: applications and considerations in measuring muscle performance. Isokin Exerc Sci 3:4–15

Stokes MJ, Dalton PA (1991) Acoustic myography for investigating human skeletal muscle fatigue. J Appl Physiol 71:1422–1426

Stout JR, Housh TJ, Johnson GO, Evetovich TK, Smith DB (1997) Mechanomyography and oxygen consumption during incremental cycle ergometry. Eur J Appl Physiol 76:363–367

Westbury JR, Shaughnessy TG (1987) Associations between spectra representation of the surface electromyogram and fiber type distribution in human masseter muscle. Electromyogr Clin Neurophysiol 27:427–435

Woods JJ, Bigland-Ritchie B (1983) Linear and non-linear surface EMG/force relationships in human muscles. Am J Phys Med 62:287–299

Yoshitake Y, Ue H, Myazaki M, Moritani T (2001) Assessment of lower-back muscle fatigue using electromyography, mechanomyography, and near-infrared spectroscopy. Eur J Appl Physiol 84:174–179

Zhang Y-T, Frank CB, Rangayyan RM, Bell GD (1992) A comparative study of simultaneous vibromyography and electromyography with active human quadriceps. IEEE Trans Biomed Eng 39:1045–1052